Proofs about Linear Maps

Existence, uniqueness

MTH 317H · Week 7 · Review + proof writing

Name: _____

Today at a glance

By the end of class you should be able to:

1. pick the right *proof framework* (first line, last line) for "injective", "surjective", "not injective", "not surjective";
2. compute $\operatorname{null} T$ and $\operatorname{range} T$ and use them to decide injectivity and surjectivity;
3. spot (and repair) the most common logical errors in proofs about functions;
4. find and verify inverses, and prove facts about compositions.

1 Warm-up: sort the maps

Here are four linear maps.

$$D\colon \mathcal{P}_3 \to \mathcal{P}_3,\quad D(p) = p', \qquad S\colon \mathbb{R}^2 \to \mathbb{R}^3,\quad S(x,y) = (x,\ y,\ x+y),$$
$$P\colon \mathbb{R}^3 \to \mathbb{R}^2,\quad P(x,y,z) = (x,\ y), \qquad E\colon \mathcal{P}_2 \to \mathbb{R}^3,\quad E(p) = \big(p(0),\ p(1),\ p(2)\big).$$

Task 1.1 (Gut feeling first)**.** Place each map in the grid with a *one-line reason.*

	injective	**not injective**
surjective	E: a nonzero $p \in \mathcal{P}_2$ has at most 2 roots, so $p(0) = p(1) = p(2) = 0$ forces $p = 0$; and every (a,b,c) is hit, by an interpolating quadratic.	P: $P(a,b,0) = (a,b)$ hits everything, but $P(0,0,1) = P(0,0,0)$, so the z-coordinate is "forgotten".
not surjective	S: (x,y) can be read off the first two coordinates of the output, but $(0,0,1)$ is missed, since every output satisfies $w_3 = w_1 + w_2$.	D: $D(1) = D(0) = 0$, and x^3 is never a derivative of a polynomial in $\mathcal{P}_3$.

Task 1.2 (Can you fill every cell?)**.** (a) Invent a linear map $\mathbb{R}^4 \to \mathbb{R}^3$ for each cell of the grid, if you can. Which cells seem impossible?

(b) Do the same for linear maps $\mathbb{R}^4 \to \mathcal{P}_3$.

Write down a conjecture. *We will come back to it in Task 4.4.*

Solution. (a) Surjective but not injective: $(x,y,z,t) \mapsto (x,y,z)$. Neither: the zero map. The two *injective* cells look impossible: four input numbers are squeezed into three, so some nonzero input should be sent to 0. (b) Bijective: $(a,b,c,d) \mapsto a + bx + cx^2 + dx^3$. Neither: the zero map. The two "mixed" cells look impossible. *Conjecture:* if $\dim V = \dim W$, then a linear map $V \to W$ is injective exactly when it is surjective. Both observations follow from the Fundamental Theorem of Linear Maps (Task 4.4).

2 Notes: two questions about every equation $T(v) = w$

Almost every question this week is a question about solving one equation

$$T(v) = w, \qquad T\colon V \to W \text{ given}, \quad w \in W \text{ a target}, \quad v \in V \text{ unknown}.$$

For a given target there are always two questions: **Existence**: is there *at least one* solution? **Uniqueness**: is there *at most one*?

Notes: definitions

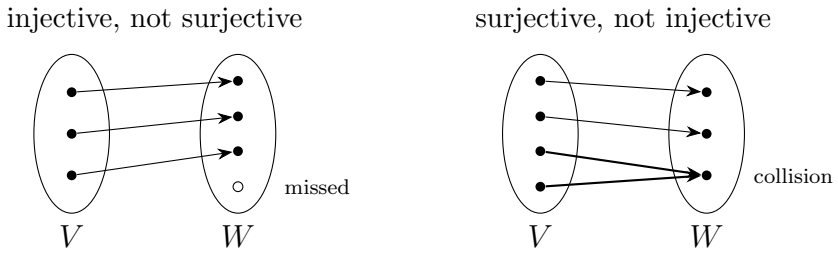
Let $T\colon V \to W$ be a function.

- T is **surjective** (onto) if for every $w \in W$ there *exists* $v \in V$ with $T(v) = w$. *(existence, for every target)*
- T is **injective** (one-to-one) if $T(u) = T(v)$ implies $u = v$. *(uniqueness, for every target)*
- T is **bijective** if it is both, so each target has *exactly one* solution. Then "solve $T(v) = w$" defines a function $W \to V$: the **inverse** T^{-1}.

If T is linear: $\operatorname{null} T = \{v \in V : T(v) = 0\}$ and $\operatorname{range} T = \{T(v) : v \in V\}$.

Key fact. A linear map T is injective $\iff \operatorname{null} T = \{0\}$. *Why:* by linearity, $T(u) = T(v) \iff T(u - v) =$ $\underline{\quad 0 \quad}$ $\iff u - v \in$ $\underline{\quad \operatorname{null} T \quad}$.

So for linear maps, uniqueness for *every* target can be checked at *one* target, $w = 0$. That is the payoff of linearity.



Notes: proof frameworks (write the first and last lines *before* you think)

The first and last lines of a proof come straight from the definitions. Write them down first, every time. The real problem-solving happens in the middle.

To prove ...	**First line**	**Last line**
T linear	Let $u, v \in V$ and $\lambda \in \mathbb{R}$.	Therefore $T(u+v) = T(u) + T(v)$ and $T(\lambda u) = \lambda\, T(u)$.
T *not* linear	Exhibit specific u, v (or λ, u) ...	... for which additivity (or homogeneity) fails. Quick test: if $T(0) \neq 0$, then T is not linear.
T injective (definition)	Let $u, v \in V$ and suppose $T(u) = T(v)$.	Therefore $u = v$.
T injective (T linear)	Let $v \in \operatorname{null} T$, i.e. $T(v) = 0$.	Therefore ___ $v = 0$ ___.
T surjective	**Let $w \in W$ be arbitrary.** *(scratch work: solve $T(v) = w$, hidden from the final proof)* Define $v := \ldots$	Then $T(v) = \cdots = w$, so $w \in \operatorname{range} T$.
T *not* injective	Exhibit specific $u \neq v$ ...	... with $T(u) = T(v)$. One example is enough.
T *not* surjective	Exhibit a specific $w \in W$ ...	... and prove $T(v) \neq w$ for **every** $v \in V$.

Connection to Lecture 7 (bases)

- Recall the proof in Lecture 7 that $(1,2,0), (2,0,0), (3,1,2)$ span $\mathbb{R}^3$: *exploration* (solve for a_1, a_2, a_3) → "Let $(x,y,z) \in \mathbb{R}^3$ be given, arbitrary" → "Define $a_1 = \ldots$" → "Then $a_1 v_1 + a_2 v_2 + a_3 v_3 = (x,y,z)$." That is *exactly* the surjectivity framework.
- The proposition "$v_1, \ldots, v_n$ is a basis $\iff$ every $v \in V$ can be written *uniquely* as a linear combination" is existence + uniqueness in disguise.

The dictionary. Given a list $v_1, \ldots, v_n$ in V, define $S\colon \mathbb{R}^n \to V$ by $S(a_1, \ldots, a_n) = a_1 v_1 + \cdots + a_n v_n$. Then

the list $v_1, \ldots, v_n$ ...	**the map S ...**	**the question**
spans V	is surjective	existence of coefficients
is linearly independent	is injective	uniqueness of coefficients
is a basis of V	is bijective	exactly one choice of coefficients

(You can prove this in Task 3.8 ★.)

3 Investigation: encoding $\mathbb{R}^4$ as cubic polynomials

Riley wants to send four numbers (a, b, c, d) to Jordan, but the only channel available transmits cubic polynomials. Riley encodes the numbers with the linear map $T\colon \mathbb{R}^4 \to \mathcal{P}_3$,

$$T\begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix} \;=\; bx^3 + (c+d)x^2 + (a-b)x - (a+d).$$

(We write (a, b, c, d) for the column vector when it fits better in a line of text.) Jordan receives a polynomial and has to *decode* it. Two questions drive this section:

- Can Jordan always decode *unambiguously*? (Is T injective?)
- Is *every* cubic polynomial a legitimate message? (Is T surjective?)

Task 3.1 (Get to know T)**.** Compute $T(e_1), T(e_2), T(e_3), T(e_4)$ and $T(1, 1, 1, 1)$. How is $T(1, 1, 1, 1)$ related to the first four? Write $T(a, b, c, d)$ as a linear combination of four *fixed* polynomials.

Solution. $T(e_1) = x-1$, $T(e_2) = x^3-x$, $T(e_3) = x^2$, $T(e_4) = x^2-1$, and $T(1, 1, 1, 1) = x^3+2x^2-2$, which is the sum of the four. In general $T(a, b, c, d) = a(x-1)+b(x^3-x)+c\,x^2+d(x^2-1)$. So T is the map S from the dictionary for the list $x-1,\ x^3-x,\ x^2,\ x^2-1$ (in particular, T is linear).

Task 3.2 (Twin hunt)**.** We know $T(1, 1, 1, 1) = x^3 + 2x^2 - 2$. Try to find a *different* $v \in \mathbb{R}^4$ with $T(v) = x^3 + 2x^2 - 2$, a "twin" of $(1, 1, 1, 1)$. Record what you tried. If you cannot find one, does that prove there is none? What exactly would a proof have to show?

Solution. There is no twin: T is injective (Tasks 3.3–3.4). Failing to find one is *evidence*, not proof, because there are infinitely many candidates. A proof must show that *every* v with $T(v) = x^3 + 2x^2 - 2$ equals $(1, 1, 1, 1)$. That is a uniqueness statement, which is what injectivity gives.

Task 3.3 (Prove it: injectivity from the definition)**.** Complete the proof. The first and last lines are given.

Proof. Let $u = (a, b, c, d)$ and $u' = (a', b', c', d')$ be in $\mathbb{R}^4$, and suppose $T(u) = T(u')$, i.e.

$$bx^3 + (c+d)x^2 + (a-b)x - (a+d) \;=\; b'x^3 + (c'+d')x^2 + (a'-b')x - (a'+d').$$

Two polynomials in $\mathcal{P}_3$ are equal exactly when **their coefficients of $1, x, x^2, x^3$ agree** (this uses the linear independence of $1, x, x^2, x^3$). Comparing coefficients,

Solution.

$$x^3\colon\ b = b', \qquad x^2\colon\ c+d = c'+d', \qquad x\colon\ a-b = a'-b', \qquad 1\colon\ a+d = a'+d'.$$

From $b = b'$ and $a-b = a'-b'$ we get $a = a'$. Then $a+d = a'+d'$ gives $d = d'$, and $c+d = c'+d'$ gives $c = c'$.

Therefore $u = u'$, and T is injective. □

Task 3.4 (Prove it again: the linear shortcut)**.** Find $\operatorname{null} T$ and conclude that T is injective. Then:

(a) Compare with Task 3.3. How is this system of equations related to the one there?

(b) Where *exactly* is linearity used when you deduce injectivity from $\operatorname{null} T = \{0\}$?

(c) Finish the twin hunt: in one sentence, explain why $(1, 1, 1, 1)$ has no twin.

Solution. Let $v = (a, b, c, d) \in \operatorname{null} T$. Then $bx^3 + (c + d)x^2 + (a - b)x - (a + d) = 0$, so $b = 0$, $c + d = 0$, $a - b = 0$, $a + d = 0$. Hence $b = 0$, $a = 0$, $d = 0$, $c = 0$, so $v = 0$. Thus $\operatorname{null} T = \{0\}$ and T is injective. (a) It is the system of 3.3 with $a - a', b - b', c - c', d - d'$ renamed a, b, c, d: the "difference" system. (b) In the step $T(u) = T(u') \Rightarrow T(u - u') = T(u) - T(u') = 0$, which uses additivity and homogeneity. (c) If $T(v) = T(1, 1, 1, 1)$, injectivity forces $v = (1, 1, 1, 1)$.

Task 3.5 (Decode a message)**.** Jordan receives $2x^3 + x^2 - 3x - 1$. Find the message $v \in \mathbb{R}^4$, and *check* your answer by computing $T(v)$.

Solution. We need $b = 2$, $c + d = 1$, $a - b = -3$, $-(a + d) = -1$. So $a = -1$, $d = 2$, $c = -1$: $v = (-1, 2, -1, 2)$. Check: $T(v) = 2x^3 + (-1 + 2)x^2 + (-1 - 2)x - (-1 + 2) = 2x^3 + x^2 - 3x - 1$. ✓

Task 3.6 (Decode *every* message: surjectivity)**.** Prove that T is surjective, following the framework.

Scratch work (not part of the final proof). Given $q = \alpha x^3 + \beta x^2 + \gamma x + \delta$, solve $T(a, b, c, d) = q$ for a, b, c, d in terms of $\alpha, \beta, \gamma, \delta$.

Solution. $b = \alpha$; $a - b = \gamma \Rightarrow a = \alpha + \gamma$; $-(a + d) = \delta \Rightarrow d = -\alpha - \gamma - \delta$; $c + d = \beta \Rightarrow c = \alpha + \beta + \gamma + \delta$.

Proof. Let $q = \alpha x^3 + \beta x^2 + \gamma x + \delta \in \mathcal{P}_3$ be arbitrary. Define $v = (_____, _____, _____, _____)$. Then $T(v) = \ldots$ (compute):

Solution. $v = (\alpha + \gamma,\ \alpha,\ \alpha + \beta + \gamma + \delta,\ -\alpha - \gamma - \delta)$, and

$$\begin{aligned} T(v) &= \alpha x^3 + \big[(\alpha + \beta + \gamma + \delta) + (-\alpha - \gamma - \delta)\big]x^2 + \big[(\alpha + \gamma) - \alpha\big]x - \big[(\alpha + \gamma) + (-\alpha - \gamma - \delta)\big] \\ &= \alpha x^3 + \beta x^2 + \gamma x + \delta. \end{aligned}$$

Hence $T(v) = q$. Since q was arbitrary, T is surjective. □

Afterwards: your formula $q \mapsto v(q)$ is a *decoder*, a function $\mathcal{P}_3 \to \mathbb{R}^4$. Test it on the message of Task 3.5. What are $T(v(q))$ and $v(T(a, b, c, d))$? (Keep this in mind for Section 6.)

Solution. With $\alpha = 2, \beta = 1, \gamma = -3, \delta = -1$: $v = (-1, 2, -1, 2)$, as in 3.5. The proof shows $T(v(q)) = q$. Substituting $\alpha = b$, $\beta = c+d$, $\gamma = a-b$, $\delta = -(a+d)$ gives $v(T(a, b, c, d)) = (a, b, c, d)$. So the decoder is T^{-1}.

Task 3.7 (What if not? Design a leaky encoder)**.** (a) Change the formula for T *a little* (one letter or one sign) so that the new map T' is *not* injective. Find a *collision*: two different messages with the same code.

(b) Is your T' surjective? Prove it, or find a polynomial that no message encodes to.

(c) Use your example to conjecture a relationship between injectivity and surjectivity for a linear map $\mathbb{R}^4 \to \mathcal{P}_3$.

Solution. Sample: change $-(a+d)$ to $-(c+d)$, so $T'(a,b,c,d) = bx^3+(c+d)x^2+(a-b)x-(c+d)$. (a) $T'(0,0,1,-1) = 0 = T'(0)$. Better still, $T'(1,1,2,0) = x^3 + 2x^2 - 2 = T'(1,1,1,1)$: now $(1,1,1,1)$ *has* a twin. (b) Not surjective: the constant coefficient of $T'(v)$ always equals minus its x^2-coefficient, so $q = 1$ is never hit. (c) This map is neither injective nor surjective. Conjecture: for linear maps between spaces of the same dimension, injective $\iff$ surjective.

Task 3.8 (★ The dictionary)**.** Let $v_1, \dots, v_n \in V$ and $S(a_1, \dots, a_n) = a_1v_1 + \cdots + a_nv_n$. Prove: (a) S is linear; (b) S is injective $\iff$ $v_1, \dots, v_n$ is linearly independent; (c) S is surjective $\iff$ $v_1, \dots, v_n$ spans V. (d) Use Task 3.1 to conclude that $x-1,\ x^3-x,\ x^2,\ x^2-1$ is a basis of $\mathcal{P}_3$.

Solution. (a) Direct check from the vector space axioms. (b) $\operatorname{null} S = \{(a_i) : \sum a_iv_i = 0\}$, and this is $\{0\}$ exactly when the only vanishing linear combination is the trivial one. (c) $\operatorname{range} S = \operatorname{span}(v_1, \dots, v_n)$, which equals V exactly when the list spans. (d) T is S for that list, and T is bijective (3.4, 3.6), so the list is linearly independent and spanning.

Takeaways

- For a linear map, to prove uniqueness for *every* target it is enough to solve $T(v) =$ __0__.
- A surjectivity proof has three moves: take an __**arbitrary**__ target; *define* a candidate (found by scratch work); __**verify**__ that it maps to the target.
- Not finding a counterexample is _**evidence**_, not __**proof**__.

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4 Investigation: squeezing $\mathbb{R}^4$ into $\mathbb{R}^3$

Now four numbers are compressed into three:

$$T\colon \mathbb{R}^4 \to \mathbb{R}^3, \qquad T\begin{pmatrix} x \\ y \\ z \\ t \end{pmatrix} = \begin{pmatrix} 2x + z - 3t \\ y - 7t \\ 6x - y + 3z - 2t \end{pmatrix}.$$

Some information has to be lost. *Where does it go, and what gets through?*

Task 4.1 (Predict, then check linearity). (a) Before computing anything: could T be injective? Could it be surjective?

(b) Prove that T is linear. *Hint:* write $T(x,y,z,t) = x(\ ) + y(\ ) + z(\ ) + t(\ )$.

(c) *Pose:* change the formula slightly so that T becomes *non*-linear, and give a specific counterexample showing that additivity or homogeneity fails.

Solution. (a) Not injective (by the 1.2 conjecture); surjectivity cannot be decided by counting alone. (b) $T(x,y,z,t) = x(2,0,6) + y(0,1,-1) + z(1,0,3) + t(-3,-7,-2)$. If $u = (x,y,z,t)$, $u' = (x',y',z',t')$ and $\lambda \in \mathbb{R}$, distributing gives $T(u+u') = T(u) + T(u')$ and $T(\lambda u) = \lambda T(u)$. (c) E.g. replace $-3t$ by $-3t+1$: then $T(0) = (1,0,0) \neq 0$, impossible for a linear map. Or replace $y - 7t$ by $y - 7t^2$: $T(0,0,0,2) = (-6,-28,-4) \neq (-6,-14,-4) = 2\,T(0,0,0,1)$.

Task 4.2 (What gets lost: $\operatorname{null} T$). (a) Find $\operatorname{null} T$ and write it as the span of a list of vectors.

(b) Is your list a basis of $\operatorname{null} T$? Justify.

(c) Use (a) to produce a *collision*: two different inputs with the same output.

(d) Find *all* v with $T(v) = T(1,1,1,1)$. Compare with the twin hunt in Section 3.

Solution. (a) $T(x,y,z,t) = 0$ means $2x + z - 3t = 0$, $y = 7t$, $6x - y + 3z - 2t = 0$. Substituting $y = 7t$ into the third equation gives $6x + 3z - 9t = 0$, which is 3 times the first: it adds no information. So $z = 3t - 2x$ and

$$\operatorname{null} T = \{(x,\ 7t,\ 3t-2x,\ t) : x,t \in \mathbb{R}\} = \operatorname{span}\big((1,0,-2,0),\ (0,7,3,1)\big).$$

(b) Yes: the list spans by (a), and it is linearly independent (look at the first and last coordinates). So $\dim \operatorname{null} T = 2$. (c) $T(1,0,-2,0) = 0 = T(0)$. (d) $T(1,1,1,1) = (0,-6,6)$, and $T(v) = T(1,1,1,1) \iff v - (1,1,1,1) \in \operatorname{null} T$. So the solutions are $(1+s,\ 1+7r,\ 1-2s+3r,\ 1+r)$, $s,r \in \mathbb{R}$: a whole plane of twins, compared with exactly one in Section 3.

Task 4.3 (What gets through: $\operatorname{range} T$). (a) Explain why $\operatorname{range} T = \operatorname{span}\big(T(e_1), T(e_2), T(e_3), T(e_4)\big)$, and compute these vectors.

(b) *The checksum.* Find numbers p, q such that *every* output $(w_1,w_2,w_3) = T(v)$ satisfies $w_3 = p\,w_1 + q\,w_2$. (Compare the third formula with the first two.)

(c) Prove $\operatorname{range} T = \{(w_1,w_2,w_3) \in \mathbb{R}^3 : w_3 = 3w_1 - w_2\}$. (Two inclusions!)

(d) Prove that T is not surjective. Your argument must rule out *every* input.

Solution. (a) $T(x,y,z,t) = xT(e_1) + yT(e_2) + zT(e_3) + tT(e_4)$, so $\operatorname{range} T \subseteq \operatorname{span}(\dots)$. Conversely each $T(e_i) \in \operatorname{range} T$, and $\operatorname{range} T$ is a subspace. The vectors are $(2,0,6)$, $(0,1,-1)$, $(1,0,3)$, $(-3,-7,-2)$. (b) $3(2x+z-3t) - (y-7t) = 6x - y + 3z - 2t$, so $p=3$, $q=-1$. (c) "$\subseteq$" is (b). "$\supseteq$": given $(w_1,w_2,3w_1-w_2)$, we have $T(0,w_2,w_1,0) = (w_1,\ w_2,\ -w_2+3w_1)$. (d) If $T(v) = (0,0,1)$ for some v, the checksum would give $1 = 3\cdot 0 - 0 = 0$, a contradiction. So $\operatorname{range} T$ is the plane $w_3 = 3w_1 - w_2$, with basis $(1,0,3), (0,1,-1)$ and $\dim \operatorname{range} T = 2$.

Task 4.4 (From data to a conjecture)**.** Complete the table, using Sections 1, 3 and 4 (for the warm-up maps, rely on your answers to Task 1.1).

map	$\dim V$	$\dim \operatorname{null} T$	$\dim \operatorname{range} T$	inj.?	surj.?
$T\colon \mathbb{R}^4 \to \mathcal{P}_3$ (Section 3)	**4**	**0**	**4**	**Y**	**Y**
$T\colon \mathbb{R}^4 \to \mathbb{R}^3$ (Section 4)	**4**	**2**	**2**	**N**	**N**
$D\colon \mathcal{P}_3 \to \mathcal{P}_3$	**4**	**1**	**3**	**N**	**N**
$S\colon \mathbb{R}^2 \to \mathbb{R}^3$	**2**	**0**	**2**	**Y**	**N**
$P\colon \mathbb{R}^3 \to \mathbb{R}^2$	**3**	**1**	**2**	**N**	**Y**
$E\colon \mathcal{P}_2 \to \mathbb{R}^3$	**3**	**0**	**3**	**Y**	**Y**

(a) Conjecture a formula relating $\dim V$, $\dim \operatorname{null} T$ and $\dim \operatorname{range} T$.

(b) Assuming your formula, explain the "impossible cells" from Task 1.2.

Solution. (a) $\dim V = \dim \operatorname{null} T + \dim \operatorname{range} T$: the *Fundamental Theorem of Linear Maps*. (b) For $T\colon \mathbb{R}^4 \to \mathbb{R}^3$: $\dim \operatorname{range} T \le 3$, so $\dim \operatorname{null} T \ge 1$ and T is never injective. For $T\colon \mathbb{R}^4 \to \mathcal{P}_3$: T injective $\iff \dim \operatorname{null} T = 0 \iff \dim \operatorname{range} T = 4 \iff \operatorname{range} T = \mathcal{P}_3$, since a subspace of full dimension is the whole space.

Homework launch: HW 5

The next two tasks get you started on HW 5. They help you plan, but they do not do the homework for you.

Task 4.5 (Null spaces in disguise)**.** Many subspaces are cut out by linear equations, which makes them null spaces. For example, the subspace from HW 5, Problem 4,

$$U = \{x \in \mathbb{R}^5 : x_1 = 2x_2 \text{ and } x_4 = 6x_5\},$$

is $\operatorname{null} L$ for the linear map $L\colon \mathbb{R}^5 \to \mathbb{R}^2$, $L(x) = (x_1 - 2x_2,\ x_4 - 6x_5)$.

(a) Check that $U = \operatorname{null} L$ and that L is surjective. Assuming the formula from Task 4.4, what must $\dim U$ be?

(b) Do the same for HW 5, Problem 5: $U = \{p \in \mathcal{P}_3 : p'(2) = 0\}$. Find a linear map $\varphi\colon \mathcal{P}_3 \to \mathbb{R}$ with $\operatorname{null} \varphi = U$. What must $\dim U$ be?

(c) How does knowing $\dim U$ in advance help you check your homework answers? *Hint:* in LADR 2.C, what can you say about a linearly independent list in U whose length is $\dim U$?

Solution. (a) $x \in \operatorname{null} L \iff x_1 - 2x_2 = 0$ and $x_4 - 6x_5 = 0 \iff x \in U$. Since $L(a, 0, 0, b, 0) = (a, b)$, L is surjective, so $\dim \operatorname{range} L = 2$ and $\dim U = 5 - 2 = 3$. (b) $\varphi(p) = p'(2)$, which is linear because differentiation and evaluation are. It is surjective, since $\varphi(cx) = c$, so $\dim U = 4 - 1 = 3$. (c) Any basis of U must have exactly 3 vectors. Moreover, a linearly independent list of length $\dim U$ in U is automatically a basis of U (LADR 2.C), so only independence would need checking.

Task 4.6 (Linear or not?)**.** For each map, predict whether it is linear, then *plan* the proof: write its first line from the framework table, or, if you think the map is not linear, the specific inputs you would use as a counterexample. Do not write the full proofs now; that is the homework.

map (HW 5, Problem 8)	**linear?**	**your plan**
(i) $\mathcal{P}_4 \to \mathcal{P}_4,\ p \mapsto x^2 p''(x)$		
(ii) $\mathcal{P}_4 \to \mathcal{P}_4,\ p \mapsto p''(x) + x^2$		
(iii) $\mathbb{R}^4 \to \mathbb{R}^3,\ x \mapsto (x_1 x_2,\ x_1,\ 5x_3 + x_4)$		
(iv) $\mathbb{R}^4 \to \mathbb{R}^3,\ x \mapsto (x_1 - 3x_2,\ x_1 + 7,\ 5x_3 + x_4)$		
(v) $\mathbb{R}^4 \to \mathbb{R}^3,\ x \mapsto (x_1 - 3x_2,\ x_1,\ 5x_3 + x_4)$		

No solution is given here: this is HW 5, Problem 8.

5 Grade the proof

Each argument below was submitted by a (fictional) student. For each one: **(1)** Verdict: is it a valid proof? **(2)** If not, point to the *first* line that fails and say why. **(3)** Repair it, or say that the claim itself is false. Read line by line and ask, every time, "*Why does this follow?*"

A. *Claim: T from Section 3 is injective.* "Since T is linear, $T(0) = 0$. So $0 \in \operatorname{null} T$, and therefore $\operatorname{null} T = \{0\}$. Hence T is injective."

> **Solution. Invalid.** "Therefore $\operatorname{null} T = \{0\}$" does not follow: $T(0) = 0$ only shows that 0 is *one* element of $\operatorname{null} T$. Every linear map satisfies $T(0) = 0$, including the non-injective T of Section 4. The claim is true, and Task 3.4 proves it.

B. *Claim: T from Section 3 is surjective.* "Let $v = (a, b, c, d) \in \mathbb{R}^4$. Then $T(v) = bx^3 + (c+d)x^2 + (a-b)x - (a+d)$ has degree at most 3, so $T(v) \in \mathcal{P}_3$. Since v was arbitrary, T is surjective."

> **Solution. Invalid.** It shows that T maps *into* $\mathcal{P}_3$, which is true of any map with codomain $\mathcal{P}_3$. The quantifiers are in the wrong order: surjectivity means "for every q there is a v", so the proof must *start* with an arbitrary $q \in \mathcal{P}_3$. Repair: Task 3.6.

C. *Claim: T from Section 3 is injective.* "Let $u, u' \in \mathbb{R}^4$ with $u = u'$. Then $T(u) = T(u')$, since T is a function. Hence T is injective."

> **Solution. Invalid.** It proves the converse, $u = u' \Rightarrow T(u) = T(u')$, which is true for *every* function. Injectivity is $T(u) = T(u') \Rightarrow u = u'$.

D. *Claim: T from Section 4 is not surjective.* "$T(e_1) = (2, 0, 6)$, $T(e_2) = (0, 1, -1)$, $T(e_3) =$

$(1, 0, 3)$, $T(e_4) = (-3, -7, -2)$. None of these is $(0, 0, 1)$, so $(0, 0, 1) \notin \operatorname{range} T$. Hence T is not surjective."

> **Solution. Invalid**, although the claim is true. The range contains all *linear combinations* of these four vectors, not just the four. The same reasoning would "prove" that $(2, 1, 5) \notin \operatorname{range} T$, yet $(2, 1, 5) = T(e_1) + T(e_2)$. Repair: the checksum argument of Task 4.3(d) rules out every input at once.

E. *Claim: T from Section 4 is not injective.* "$T(1, 0, -2, 0) = (2 - 2,\ 0,\ 6 - 6) = (0, 0, 0) = T(0, 0, 0, 0)$, and $(1, 0, -2, 0) \neq (0, 0, 0, 0)$. So T is not injective."

> **Solution. Valid.** To disprove a "for all" statement, one counterexample suffices. The line "$(1, 0, -2, 0) \neq 0$" is essential, because a collision needs two *different* inputs.

Task 5.1 (★ Turn the tables)**.** Write a short "proof" about one of today's maps that *looks* convincing but contains one subtle error. Identify the error and explain how to correct it.

Classic traps

1. $T(0) = 0$ holds for *every* linear map. It says $0 \in \operatorname{null} T$, not $\operatorname{null} T = \{0\}$.
2. Mind the quantifier order. Surjective means *for every w there is a v*, so start from w.
3. Don't prove the converse. Injective is $T(u) = T(u') \Rightarrow u = u'$, not the other way around.
4. Finitely many checks can *disprove* a "for all" statement, but they can never prove a "there is no ..." statement.

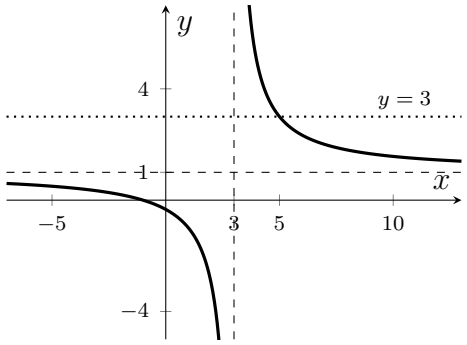
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6 Inverses and compositions (Week 6, Problems 5 and 7)

In Task 3.6 you built a decoder that undoes T. In Week 6 you proved that $f\colon X \to Y$ *is bijective if and only if it has an inverse* $h\colon Y \to X$, meaning $h(f(x)) = x$ for all $x \in X$ and $f(h(y)) = y$ for all $y \in Y$. The two problems below involve functions that need not be linear. The frameworks from Section 2 still apply word for word.

An inverse with excluded values

$$r\colon \mathbb{R} \setminus \{3\} \longrightarrow \mathbb{R} \setminus \{1\}, \qquad r(x) = \frac{x + 1}{x - 3}.$$



Task 6.1 (Look before you prove)**.** (a) Which horizontal lines $y = c$ meet the graph exactly once? Which meet it not at all?

(b) What does this "horizontal line test" have to do with injectivity and surjectivity?

(c) Why is the codomain $\mathbb{R} \setminus \{1\}$ rather than $\mathbb{R}$?

Solution. (a) Every line $y = c$ with $c \neq 1$ meets the graph exactly once; $y = 1$ (the horizontal asymptote) never does. (b) "At most once" for every c in the codomain is injectivity; "at least once" is surjectivity. (c) The value 1 is never attained. With codomain $\mathbb{R}$, r would not be surjective.

Task 6.2 (Prove it)**.** (a) Check that the formula defines a function $\mathbb{R} \setminus \{3\} \to \mathbb{R} \setminus \{1\}$. There are *two* things to check.

(b) *Scratch work:* solve $y = r(x)$ for x. Use the result to define a candidate inverse s, stating its domain and codomain.

(c) Check that s takes values in its claimed codomain, and verify both compositions. Conclude that r is bijective.

Solution. (a) For $x \neq 3$ the denominator is nonzero, so $r(x)$ is defined. If $r(x) = 1$, then $x + 1 = x - 3$, i.e. $1 = -3$, a contradiction. So $r(x) \in \mathbb{R} \setminus \{1\}$. (b) $y = \frac{x+1}{x-3} \Rightarrow yx - 3y = x + 1 \Rightarrow x(y-1) = 3y + 1$. Candidate: $s\colon \mathbb{R} \setminus \{1\} \to \mathbb{R} \setminus \{3\}$, $s(y) = \frac{3y+1}{y-1}$. (c) For $y \neq 1$, $s(y)$ is defined and $s(y) - 3 = \frac{4}{y-1} \neq 0$, so $s(y) \neq 3$. Since $s(y) + 1 = \frac{4y}{y-1}$, we get $r(s(y)) = \frac{4y/(y-1)}{4/(y-1)} = y$. For $x \neq 3$, $3r(x) + 1 = \frac{4x}{x-3}$ and $r(x) - 1 = \frac{4}{x-3}$, so $s(r(x)) = x$. Hence s is an inverse of r, so r is bijective (Week 6, Problem 3) and $r^{-1} = s$.

Task 6.3 (What if not?)**.** (a) Keep the formula, but change the codomain to $\mathbb{R}$. Which of injective, surjective, bijective survive?

(b) For $r_2(x) = \dfrac{2x+1}{x-3}$, which value must be removed from the codomain? Conjecture the excluded output for $\dfrac{ax+b}{cx+d}$ with $c \neq 0$.

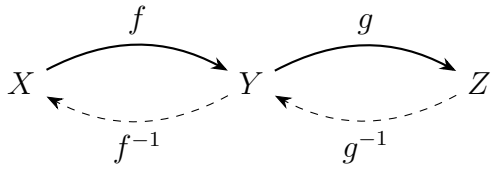
(c) ★ Try $\dfrac{2x+6}{x+3}$. What goes wrong? Find a condition on a, b, c, d that rules this out.

(d) ★ Write the coefficients of r and of your s as 2×2 matrices $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and multiply them. Notice anything?

Solution. (a) Still injective; no longer surjective (1 is missed); so not bijective. Surjectivity depends on the codomain. (b) 2. In general, a/c: setting $y = a/c$ in $x(cy - a) = b - dy$ gives $0 = (bc - ad)/c$, which has no solution when $ad - bc \neq 0$. (c) $\frac{2x+6}{x+3} = 2$ for all $x \neq -3$: the function

is constant, so badly non-injective. The condition is $ad - bc \neq 0$. (d) $\left(\begin{smallmatrix} 1 & 1 \\ 1 & -3 \end{smallmatrix}\right)\left(\begin{smallmatrix} 3 & 1 \\ 1 & -1 \end{smallmatrix}\right) = \left(\begin{smallmatrix} 4 & 0 \\ 0 & 4 \end{smallmatrix}\right) = 4I$. Composing such fractions corresponds to multiplying matrices (up to a scalar), and $ad - bc$ is the determinant.

Composition of bijections



Task 6.4 (Socks and shoes)**.** (a) *Warm-up.* Let $f, g\colon \mathbb{R} \to \mathbb{R}$, $f(x) = x + 1$, $g(x) = 2x$. Compute $g \circ f$ and $(g \circ f)^{-1}$. Which one equals $(g \circ f)^{-1}$: $g^{-1} \circ f^{-1}$ or $f^{-1} \circ g^{-1}$? Explain using socks and shoes.

(b) Now let $f\colon X \to Y$ and $g\colon Y \to Z$ be bijective. Prove directly that $g \circ f$ is injective. *(Which row of the framework table are you in?)*

(c) Prove directly that $g \circ f$ is surjective.

(d) Prove that $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$ by checking both compositions.

Solution. (a) $(g \circ f)(x) = 2x + 2$ and $(g \circ f)^{-1}(y) = \frac{y}{2} - 1$. Also $f^{-1}(y) = y - 1$, $g^{-1}(y) = \frac{y}{2}$, so $(f^{-1} \circ g^{-1})(y) = \frac{y}{2} - 1$ ✓, whereas $(g^{-1} \circ f^{-1})(y) = \frac{y-1}{2}$ ×. Socks go on first, then shoes; to undo this you take off the shoes first, then the socks. (b) Suppose $g(f(x_1)) = g(f(x_2))$. Injectivity of g gives $f(x_1) = f(x_2)$, and injectivity of f gives $x_1 = x_2$. (c) Let $z \in Z$ be arbitrary. Since g is surjective, there is $y \in Y$ with $g(y) = z$. Since f is surjective, there is $x \in X$ with $f(x) = y$. Then $(g \circ f)(x) = z$. (d) With $h = f^{-1} \circ g^{-1}\colon Z \to X$: $h((g \circ f)(x)) = f^{-1}(g^{-1}(g(f(x)))) = f^{-1}(f(x)) = x$ and $(g \circ f)(h(z)) = g(f(f^{-1}(g^{-1}(z)))) = g(g^{-1}(z)) = z$. So $h = (g \circ f)^{-1}$. The domains also force the order: $Z \xrightarrow{g^{-1}} Y \xrightarrow{f^{-1}} X$.

Task 6.5 (What if not? Converses)**.** Suppose we know only that $g \circ f$ is bijective. Must f be injective? Surjective? Must g be injective? Surjective? Make four conjectures, then prove each one or find a counterexample using small finite sets and arrow diagrams.

Solution. f must be injective: if $f(x_1) = f(x_2)$, then $g(f(x_1)) = g(f(x_2))$, so $x_1 = x_2$. g must be surjective: given z, pick x with $g(f(x)) = z$; then $y = f(x)$ works. The other two can fail. Take $X = Z = \{1\}$, $Y = \{1, 2\}$, $f(1) = 1$, $g(1) = g(2) = 1$. Then $g \circ f$ is bijective, but f is not surjective and g is not injective.

Task 6.6 (★ Back to linear maps)**.** Let $S\colon U \to V$ and $T\colon V \to W$ be linear and injective. Give a two-line proof, using null spaces, that $T \circ S$ is injective.

Solution. If $T(S(u)) = 0$, then $S(u) \in \operatorname{null} T = \{0\}$, so $S(u) = 0$; then $u \in \operatorname{null} S = \{0\}$. Thus $\operatorname{null}(T \circ S) = \{0\}$.

7 Exit ticket (tear off and hand in)

Name: _____

1. To prove that $T\colon V \to W$ is *surjective*, my first line is **Let $w \in W$ be arbitrary.** and my last line is **Therefore $T(v) = w$ (for the v I defined).**

2. To prove that T is *not* surjective, I must **find one $w \in W$ and show $T(v) \neq w$ for every $v \in V$.**

3. Rate your confidence, 1 (lost) to 4 (I could teach it):

proving a map is injective	1	2	3	4
proving a map is surjective	1	2	3	4
computing $\operatorname{null} T$ and $\operatorname{range} T$	1	2	3	4
finding and verifying an inverse	1	2	3	4

4. **Pose a problem.** Write one question about today's ideas that you would like answered, *or* a problem you would put on the exam.